# Supplementary material for
# *A spectral theory for turbulent burning velocity across scales*

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This document accompanies the paper. Sections, figures, tables and equations numbered with the prefix S belong to this document; all other references are to the paper. The figures are produced by the notebook `make_figures.ipynb` from the same data as the figures of the paper.

## S1 Distribution of the wrinkling exponent

The paper connects its burning-velocity relation with the power-law wrinkling factor of Charlette *et al.* (2002*a*) for an exponent $\beta_w = 1/2$. Charlette *et al.* (2002*b*) determined $\beta_w$ locally with a dynamic procedure, and figure S1 shows the probability density function (PDF) of the exponents they obtained, digitised from figure 5 of their paper. The digitised curve, which ends at $\beta_w = 1$, was normalised to unit area and its moments were computed by trapezoidal integration. The mean is 0.587 and the standard deviation 0.104 (variance 0.0109); the median is 0.57 and the interquartile range 0.51–0.65. About 22 % of the distribution lies below $\beta_w = 0.5$, and the digitised PDF falls to zero at $\beta_w = 0.4$, above the fractal value $\beta_w = 1/3$ that would recover Damköhler's large-$u'$ limit $s_T \propto u'$. The distribution therefore supports an exponent close to $1/2$, centred slightly above it.

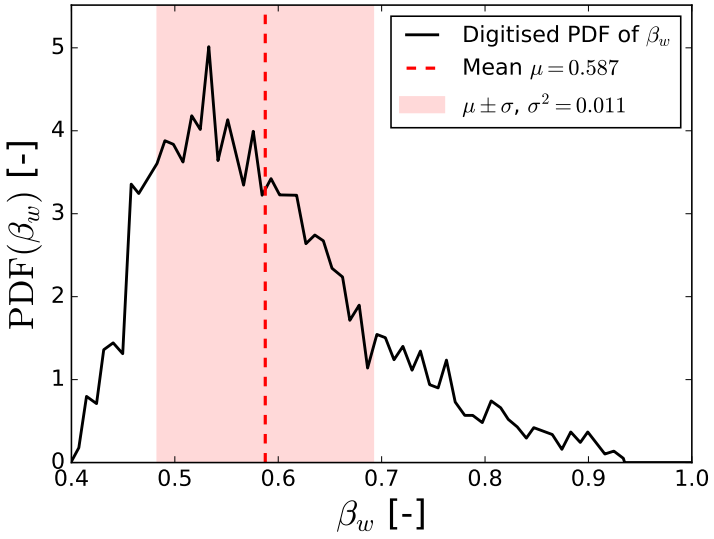



Figure S1: PDF of the local wrinkling exponent $\beta_w$ obtained with the dynamic procedure of Charlette *et al.* (2002*b*) (digitised). Dashed line, mean $\mu = 0.587$; shading, $\mu \pm \sigma$, where $\sigma^2 = 0.011$ is the variance.

## S2 Selection and treatment of the experimental data

Measured turbulent burning velocities depend on the flame configuration and on how $s_T$ is defined (Driscoll, 2008), so the datasets of table 1 of the paper were chosen to keep both as uniform as possible.

## S2.1 Flame configuration

Driscoll (2008) groups premixed turbulent flames into envelope flames (on two-dimensional slot and axisymmetric Bunsen burners), oblique flames, unattached flat flames (on low-swirl, diffuser or counterflow burners) and spherical flames (in fan-stirred vessels). All of the datasets except Aachen are envelope flames: the Michigan (2005) flames are on a slot burner, Damköhler's on tubes and the others on round Bunsen burners. The Tohoku burner has a hydrogen diffusion flame around the nozzle exit as a pilot (Kobayashi *et al.*, 1996), and the Michigan (2005) slot, 25.4 mm by 50.8 mm, is flanked by two burners whose grid-stabilised flames supply products at the same velocity (Filatyev *et al.*, 2005). The Aachen flames, on a low-swirl burner, are unattached flames in this classification.

## S2.2 Turbulent burning velocity

In an envelope flame, all of the reactants supplied by the burner pass through the flame, so mass conservation gives a mean, or global, consumption speed,

$$s_T = \frac{\dot{m}_R}{\rho_R \bar{A}}, \tag{S1}$$

where $\dot{m}_R$ is the mass flow rate of reactants, $\rho_R$ their density and $\bar{A}$ the area of a mean flame surface (Filatyev *et al.*, 2005; Driscoll, 2008). Where the source reports it, $\bar{A}$ is taken on the isosurface $\overline{\Lambda} = 0.5$ of the mean progress variable (table 1 of the paper); for Ottawa it is $\overline{\Lambda} = 0.05$, and for Michigan (2005) it is the time-averaged flame position. For Tohoku, Kobayashi *et al.* (1996) used the flame angle instead, $s_T = U \sin(\theta/2)$, where $U$ is the mean exit velocity and $\theta$ the apex angle of the mean flame cone, which they suggested lies near $\overline{\Lambda} = 0.5$. Damköhler (1940) computed flame speeds $w_{f,\max}$ and $w_{f,\min}$ from the areas of the inner and outer boundaries of the flame zone, respectively; $w_{f,\min}$ agreed with the laminar flame speed in all but two of his tests, and he used $w_{f,\max}/w_{f,\min}$ as the measure of the increase due to turbulence, which is the $s_T/s_L$ used here. Differences in equivalence ratio, within and between datasets, are accounted for by normalising $s_T$ by the laminar burning velocity $s_L$ of each mixture.

Spherically expanding flames, such as those studied at Leeds (Bradley *et al.*, 2003; Lawes *et al.*, 2012) and Princeton (Chaudhuri *et al.*, 2012; Wu *et al.*, 2015), are characterised instead by a propagation or displacement speed derived from the growth rate $\mathrm{d}R/\mathrm{d}t$ of a mean flame radius $R$, often multiplied by the ratio of burned to unburned gas density. This speed increases with $R$, because a growing flame kernel is wrinkled by an increasing part of the turbulence spectrum (Abdel-Gayed *et al.*, 1987; Bradley *et al.*, 2003; Lawes *et al.*, 2012), and Chaudhuri *et al.* (2012) correlated it with a Reynolds number based on the mean flame radius rather than on $\ell_0$. It measures a different quantity from (S1), and such data are not included.

## S2.3 Approaching turbulence

The approaching turbulence differs between the configurations. In an envelope flame, it is generated in the burner tube or nozzle, typically by a perforated or slotted plate or a grid, and in Damköhler's experiments it is the turbulence of the tube flow itself, so its spectrum is that of a turbulent pipe or nozzle flow. Pope (2000, figure 6.14) derived the model spectrum used in the paper with one-dimensional spectra measured in a range of flows, among them turbulent pipe flow and round jets. In fan-stirred vessels, the turbulence is produced by fans and is close to isotropic near the centre; its spectrum has been characterised for the Leeds vessel by Abdel-Gayed *et al.* (1987), who measured its power spectral density, and in Walker (2019).

## S2.4 Mixtures and operating conditions

To exclude differential diffusion, all of the datasets except Damköhler's are methane–air flames, for which the Lewis number $Le \approx 1$ (e.g. Kobayashi *et al.*, 1996), at 0.1 MPa and about 300 K; for this reason, the

high-pressure measurements of Kobayashi *et al.* (1996) and the propane–air flames of Yuen & Gülder (2013) are not used. Damköhler's flames burned a propane–butane mixture with oxygen.

## S2.5 Laminar flame thickness

The sources define the laminar flame thickness differently. Michigan (2017) and Toronto (2013) give thermal thicknesses from detailed-chemistry calculations of the laminar flame (0.36 mm, and 0.45–1.0 mm for the flames used here); Michigan (2005) uses $\delta_L = 7.4\, D_{\mathrm{th}}/s_L = 0.35$ mm, where $D_{\mathrm{th}}$ is the thermal diffusivity of the reactants; Aachen uses $D/s_L = 0.35$ mm with the diffusivity $D$ evaluated at 1500 K; and the thicknesses of Tohoku and Toronto (2014), 0.066 mm and 0.055–0.11 mm, correspond to $D_{\mathrm{th}}/s_L$; Kobayashi *et al.* (1996) report no thickness, so the Tohoku value is computed here. Ottawa gives 0.033–0.065 mm. The paper takes $\ell_0/\delta_L$ as given by each source, or as computed from the given $\ell_0$ and $\delta_L$. With unity Schmidt number, a consistent choice would be $\delta_L = \nu/s_L$, so that $Re = u'\ell_0/\nu$. Reynolds numbers defined in this way, reported by the sources or computed from their tabulated $\nu$ and $s_L$, exceed those used in the paper by factors of 5.5, 6.6 and 8.9 for Michigan (2017), Toronto (2013) and Michigan (2005), and by 1.4 for Tohoku and Toronto (2014). The thickness enters the prediction only through $\mathcal{F}(Re)$, which varies slowly: halving or doubling $\delta_L$ changes the predicted $s_T$ by about 10 % at the median and by at most 21 %. Re-evaluating $Re$ as $u'\ell_0/\nu$ for these five datasets changes the fraction of points within a factor of 1.5 of the prediction from 71 % to 69 %, and the r.m.s. logarithmic deviation from 0.38 to 0.37, although individual datasets shift: the geometric-mean ratio of measurement to prediction changes from 1.06 to 0.96 for Michigan (2017), from 1.06 to 0.74 for Toronto (2013) and from 1.37 to 1.04 for Michigan (2005).

Damköhler (1940) did not measure $u'$ or $\ell_0$, so for his flames they are estimated from the tube flow. The integral scale is taken as his maximum mixing length in each tube, 0.11, 0.17 and 0.21 mm for the tubes of 1.385, 2.18 and 2.718 mm diameter, and the flame thickness as $\delta_L = 0.01$ mm, so that $\ell_0/\delta_L = 11$, 17 and 21; Damköhler gives a thickness of 0.1 mm for the luminous zone of his laminar flames. The turbulence Reynolds number is taken as $Re = (Re_d - 2300)^{1/2}$, where $Re_d$ is the tube Reynolds number and 2300 its laminar limit, so that $u'/s_L = (Re_d - 2300)^{1/2}/(\ell_0/\delta_L)$, and $u' = 0$ for $Re_d < 2300$.

Laminar flames were investigated at Damköhler's conditions. Freely propagating, adiabatic flames, taken as 85 % propane and 15 % butane by volume, with oxygen at 291 K and 0.1 MPa were computed for each of the 22 mixtures with `Cantera` (Goodwin *et al.*, 2025) and two kinetic mechanisms, UCSD mech[1] and USC mech II (Wang *et al.*, 2007). For his laminar tests, the median ratio of $w_{f,\min}$ to the computed $s_L$ is 1.01 with UCSD mech and 1.11 with USC mech II. The thermal thickness of these flames, $(T_b - T_u)/\max(\mathrm{d}T/\mathrm{d}x)$, is 0.07–0.29 mm, comparable to the 0.1 mm of the luminous zone, whereas $\nu_u/s_L$, with the kinematic viscosity $\nu_u$ of the reactants, is 0.004–0.019 mm. The thickness $\delta_L = 0.01$ mm used here therefore corresponds to $\nu/s_L$, rather than to a thermal thickness. The calculation is in the notebook `data/source/Damkohler_laminar_flames.ipynb` of the paper's repository.

## S2.6 Regime diagrams

Figure S2 places the datasets, with the same $u'/s_L$ and $\ell_0/\delta_L$ as figure 1(*a*) of the paper, on two variants of the regime diagram.

Skiba *et al.* (2018) measured preheat- and reaction-layer thicknesses in the Michigan flames and compared them with flame structures reported in earlier experiments and simulations. The Klimov–Williams criterion $Ka = 1$ did not separate thin flamelets from flames with broadened preheat zones, so they replaced it by a ratio of turbulent to molecular diffusivity, $D_T/D^* = u'\ell_0/(s_L\delta_L)$, which equals $Re$. Flames with $D_T/D^* \gtrsim 180$ had broadened preheat zones but thin reaction layers, and no broadened or broken reaction layers were observed up to $Ka = 533$. Their measured regime diagram (figure S2*a*) adds tentative boundaries at $Ka = 60$, which extends the broadened-preheat regime to small $\ell_0/\delta_L$, at $Ka = 1$ for $D_T/D^* > 180$, and at $Ka \approx 550$ for broadened reaction zones. Most of the present measurements

[1]Chemical-kinetic mechanisms for combustion applications, Mechanical and Aerospace Engineering (Combustion Research), University of California at San Diego, https://web.eng.ucsd.edu/mae/groups/combustion/mechanism.html.

fall in the thin-flamelet region; the broadened-preheat region contains the high-$Ka$ Michigan (2017) and Toronto (2013) flames and part of the Aachen data, and no dataset reaches $Ka = 550$. Skiba *et al.* (2018) use the diffusive thickness of Peters (2000), with the diffusivity evaluated near the reaction layer, so the positions of datasets based on other definitions of $\delta_L$ (§ S2.5) are approximate.

The Bray number (Veynante *et al.*, 1997),

$$N_B = \frac{\tau s_L}{2\alpha u'}, \tag{S2}$$

compares the gas expansion across the flame, through the heat-release factor $\tau = \rho_u/\rho_b - 1$, with the turbulent velocity fluctuations; $\alpha$ is an efficiency of order unity. For $N_B > 1$, the expansion dominates and the turbulent scalar flux is counter-gradient, and for $N_B < 1$, it is gradient (Veynante *et al.*, 1997). A counter-gradient flux implies a negative turbulent diffusivity, and Nivarti & Cant (2017) found that this limits the validity of Damköhler's second hypothesis, on which the diffusive reading of the paper's relation rests. Figure S2($b$) shows $N_B = 1$ on the regime diagram of figure 1($a$) of the paper for $\tau = 6.67$, the value for the stoichiometric methane–air flames of Nivarti & Cant (2017). The efficiency is either $\alpha = 0.25$, which reproduced the transition in their simulations at $\ell_0/\delta_L = 2.75$, or increases logarithmically with $\ell_0/\delta_L$ as $\alpha = 0.4 \log_{10}(\ell_0/\delta_L) + 0.059$ (but not below 0.05), which passes close to $\alpha \approx 0.3$ at $\ell_0/\delta_L = 4$ (Kheirkhah & Gülder, 2015) and $\alpha = 0.5$ at $\ell_0/\delta_L = 11$ (Veynante *et al.*, 1997). With $\alpha = 0.25$, 17 of the 75 measurements with $u' > 0$ have $N_B < 1$, 13 of them from Michigan (2017), and the geometric-mean ratio of measured to predicted $s_T$ is 1.12 on both sides of the line. With the logarithmic $\alpha$, 29 have $N_B < 1$, and the ratios are 1.06 for $N_B < 1$ and 1.15 for $N_B > 1$.

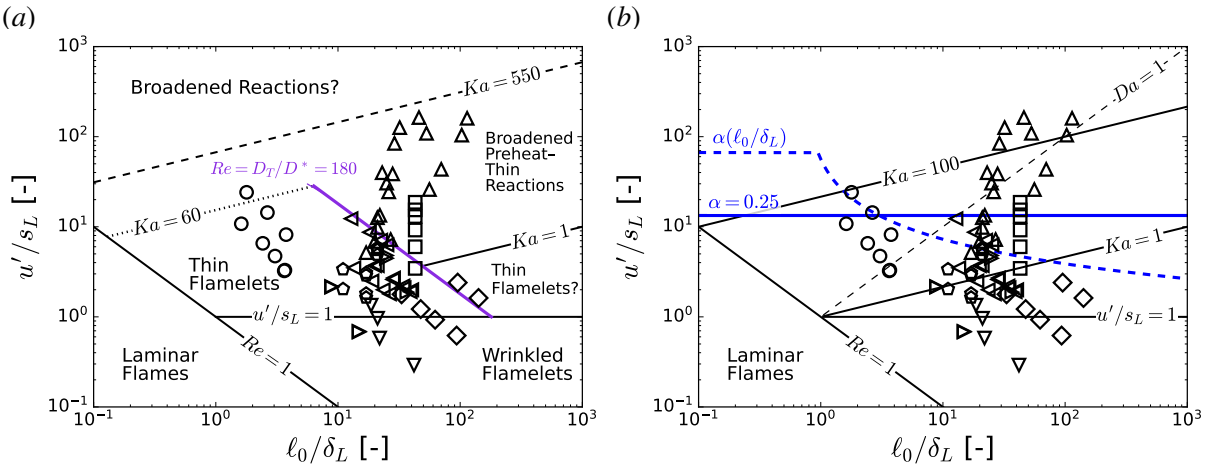



Figure S2: The datasets of table 1 of the paper (symbols as there) on ($a$) the measured regime diagram of Skiba *et al.* (2018), with their measured boundary $D_T/D^* = 180$ (purple) and tentative boundaries $Ka = 1$, 60 and 550; and ($b$) the regime diagram of figure 1($a$) of the paper with the line $N_B = 1$ for $\tau = 6.67$ and $\alpha = 0.25$ (solid blue) or the logarithmic $\alpha(\ell_0/\delta_L)$ (dashed blue). Below the blue lines, $N_B > 1$ and the scalar flux is counter-gradient.

## S3 Scaling of the burning velocity with Reynolds and Karlovitz numbers

Figure S3 shows the measured $s_T/s_L$ divided by $Ka$ against $Re$ for the eight datasets of table 1 of the paper, with the same values and symbols. The dashed lines are $100\,Re^{-n}$, labelled with their slope $-n$. Since $Ka = (u'/s_L)^{3/2}(\ell_0/\delta_L)^{-1/2}$ and $Re = (u'/s_L)(\ell_0/\delta_L)$, a series of measurements at fixed $\ell_0/\delta_L$ in which $s_T/s_L \propto (u'/s_L)^m$ follows

$$\frac{s_T/s_L}{Ka} \propto Re^{m-3/2} \tag{S3}$$

in this plane. A slope of $-1/2$ thus corresponds to Damköhler's linear scaling, $m = 1$, and a slope of $-1$ to $m = 1/2$, the strong-turbulence limit of the paper's relation $s_T/s_L = [1 + C\,\mathcal{F}(Re)\,u'/s_L]^{1/2}$ apart

from the weak variation of $\mathcal{F}$ with $Re$. Datasets in which $\ell_0/\delta_L$ varies together with $u'/s_L$ do not follow (S3) exactly.

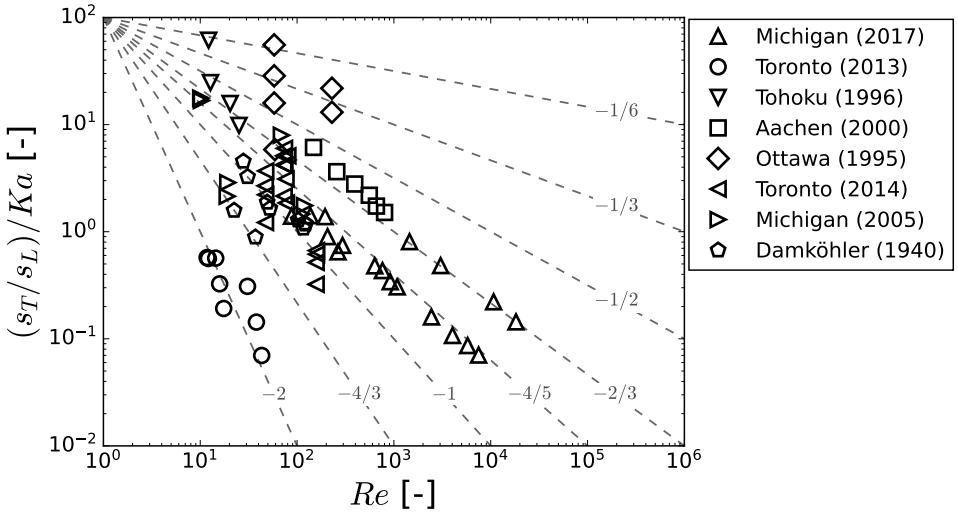



Figure S3: Measured $s_T/s_L$ divided by $Ka$ against $Re$ for the datasets of table 1 of the paper (symbols as there). Dashed lines, $100\,Re^{-n}$, labelled with their slope $-n$.

## S4 Fits to the spectral eddy viscosity and diffusivity

The functions $\nu_e^+(\kappa/\kappa_c)$ and $D_e^+(\kappa/\kappa_c)$ computed by Chollet (1984) were digitised for $0.01 \leqslant \kappa/\kappa_c \leqslant 1$ and fitted by nonlinear least squares with the form that Pope (2000) used for $\nu_e^+$,

$$\nu_e^+,\ D_e^+ = C^{-3/2}\left[A + B\exp\left(-c\,\frac{\kappa_c}{\kappa}\right)\right], \qquad C = 1.5. \tag{S4}$$

Table S1 lists the coefficients and the values at the cusp, $\kappa = \kappa_c$, which are the values used in the paper; the turbulent Schmidt number is $Sc_T = \nu_e^+/D_e^+$. The fitted curves are shown in figure 2($b$) of the paper. The regression of Pope (2000) for $\nu_e^+$ gives a cusp value 26 % lower than the fit to the curve of Chollet (1984).

| | $A$ | $B$ | $c$ | Value at $\kappa = \kappa_c$ |
|---|---|---|---|---|
| $\nu_e^+$, Pope (2000) | 0.441 | 15.2 | 3.03 | 0.640 |
| $\nu_e^+$, fit to Chollet (1984) | 0.4927 | 162.7 | 4.996 | 0.868 |
| $D_e^+$, fit to Chollet (1984) | 0.8717 | 949.5 | 5.908 | 1.88 |
| $Sc_T = \nu_e^+/D_e^+$ | | | | 0.462 |

Table S1: Coefficients of (S4) and values at the cusp.

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